Week 1 – Workshop
Gradients, Curvature, and Potential-Field Intuition

Geophysics 3A: Potential Fields & Geothermics

July 24, 2026

Duration: 3 hours **Setting:** Workshop room

Preparation

Pre-reading: Course Notes Ch. 1, 3–4
Lecture videos:
Notes:

Tips

- Work in mixed-background groups; geologist + physicist + ...
- Share your expertise freely and use each other's knowledge to fill gaps in your own.
- Work colletively and constructively towards the common aims. Value each other's opinions, feel free to respectively discuss differences of ideas.

Purpose

The purpose of this workshop is to build an intuitive and practical understanding of gradients and Laplacians as tools for interpreting geophysical fields. Using familiar examples from topography, you will connect physical intuition (slope, steepness, direction, and curvature) to their mathematical descriptions (partial derivatives, gradients, and the Laplacian), and begin to see how these ideas extend to abstract potential fields used in geophysics.

This workshop also introduces back-of-the-envelope (BOTE) reasoning as a core skill for estimating physical quantities, assessing plausibility, and developing quantitative intuition before applying more formal calculations or computational methods.

Learning Outcomes

By the end of this workshop, you should be able to:

- Describe the gradient of a scalar field in both intuitive (slope and direction) and mathematical (partial derivatives and vector representation) terms.
- Interpret maps and surfaces by identifying regions of high gradient magnitude and explaining their physical significance.
- Conceptually explain curvature (Laplacian) as changes in slope and identify where it is large or near zero in maps or surfaces.
- Construct and interpret gradient vectors graphically from contour maps.
- Apply simple back-of-the-envelope calculations to estimate physical quantities and justify assumptions using order-of-magnitude reasoning.
- Connect the concepts of gradients and curvature to the behaviour of potential fields and their role in geophysical interpretation.

Session Flow (165 min)

0–15 min: Course introduction

What the course covers, weekly rhythm, assessment structure, expectations, office hours, and where to find materials. Download mateials while instructor goes over outline of course.

15–60 min: Activity 1.A: Tectonic Plate vs. Freight Train

Work in groups of 3–4 on the back-of-the-envelope style problem.

To estimate these parameters, you will need to know/learn a few things about the structure of Earth, the physical properties of common rock types within each layer, and make assumptions (understand ranges) for required values. You can refer to the *Geology for Geophysicists* notes. You will need to keep track of units and become comfortable with uncertainty and order-of-magnitude estimation.

60–80 min: Mini-lecture and Q & A

Address questions from lecture; otherwise give a concise teaser: scalar vs vector fields, ∇h as direction of steepest increase, and ∇^2 as curvature, vector field = gradient of scalar potential ($F = \nabla\phi$).

80–85 min: Break

85–100 min: Activity 1.B: Sheet Model of Potential Fields

Hands-on demonstration of a scalar field using a stretched sheet. Boundary heights are fixed (Dirichlet boundary conditions), while pushing up/down on the sheet introduces localized sources and sinks.

Focus on how the gradient (slope) and curvature (Laplacian) respond to boundary conditions and internal forcing, and how increasing observation distance smooths the field.

100–130 min: Activity 1.C: From Elevation to Gradients

Work in groups of 3–4 on the gradient activity.

130–145 min: Activity Debrief

Discuss the answers to the Gradient Activity.

145–155 min: Wrap-up

What is a gradient and why does it matter in geophysics?

Gradients:

1. describe how fields change in space;
2. yield the direction of steepest increase;
3. their magnitude tells us how fast the field is changing; and
4. they relate a potential to a vector field, $\mathbf{F} = -\boldsymbol{\nabla}\phi$, with the sign being convention.

155–165 min: Preview: Physical properties

Next week, we'll examine what I like to call the *geophysical fudge factors*: the physical properties of Earth materials. These properties control how fields are generated, modified, and transmitted. Mathematically, they often appear as scale factors that determine magnitudes and ensure the units make sense—but physically, *properties contain the geology.*

As geophysicists, we measure fields but interpret physical properties, so it is essential to understand how these properties vary with composition and state variables such as pressure and temperature.

We'll also look at how to estimate rock properties using different approaches, including deriving bulk properties from mineralogy. This approach introduces mixing models, and we'll discuss when different mixing relationships are appropriate.

Workshop Notes

Workshop Notes

Activity 1.A
Tectonic Plate vs. Freight Train

Purpose

The purpose of this activity is both a warm-up and an introduction to back-of-the-envelope (BOTE) problem solving. So called a BOTE calculation because it is one that can be done relatively simply on a small piece of scrap paper. Such calculations often result in order-of-magnitude estimates, are a way to place rough bounds on a value, and can be used test simple hypotheses before more complex computations are required. It is a useful skill because it can be done in the field to do simple interpretations of data or during a chat with colleagues over drinks/dinner.

Learning Outcomes

By the end of this activity, you should be able to:

- Apply a systematic back-of-the-envelope strategy to estimate unknown physical quantities.
- Use order-of-magnitude reasoning and scientific notation to simplify calculations.
- Identify and justify assumptions when estimating quantities for which precise values are unavailable.
- Recognise that physical intuition—not numerical precision—is the primary goal of BOTE estimates.

How to do BOTE calculations

Work backwards. Start with the last equation you will need to solve the problem. Write down what you know and identify what you don't. Often you will need to do multiple calculations on the way to solving the problem.

A couple things to keep in mind:

round quantities – we are generally looking for order-of-magnitude estimates—calculations you can do in your head—not precise answers;

estimate unknowns – few values are truely unknown, bounds even rough can often be set;

use geometric means – you will find that when using bounds values tend towards the smaller bound, a geometic mean $g = (\text{lower} \times \text{upper})^{1/2}$ of the bounds is often closer to the truth than an arithmetic mean;

use scientific notation – quicker for many unit conversions, multiplication and division;

dimensional analyis – always check your units, it can reduce the number of mistakes and can be a way to construct equations when you can't remember the exact formula

0.1 Rounding

We can round numbers to make calculations quicker and easier. For example, how many seconds are there in a year? The exact answer requires the number of seconds in a minute, number of minutes in an hour, number of hours in a day, and number of days in a year (don't forget leap years). That's too complicated. The number is approximately 3.15×10^7 s.

There are two values I keep in my head, 3×10^7 and $\pi \times 10^7$ s. The former is simple and rounded to one digit whereas the latter often useful for canceling π, which shows up in a lot of equations.

0.2 Estimating Unknowns

Even when we have little information about the value for a property or a quantity, we can develop an estimate or place some rough bounds. It is rare that we truely have no idea.

Consider the velocity of a tectonic plate, which you will need for the eventual questions. We know the velocity of a tectonic plate is between 0 and we can easily outwalk it, so we know it is less than m s^{-1}. We can do better. Let's look at 3 ways to estimate it:

Case 1: Start with a random fact you might know.
You might have heard that it is about the rate that fingernails grow. Personally, I can't stand my fingernails to get long. I cut about a millimeter off every two weeks. There are 26 fortnights in a year so that's a velocity of 26 mm a.

Case 2: Estimate it from something that is easy to determine.
Africa and North America split from Pangea about 180 Ma. Looking at a map, I can estimate about 60° between Morocco and Florida. At the equator (close enough) the distance per degree is ~111.2 km. So there is about 6600 km between the two countries today. That would be an average rate of approximately (6600 km)/(180 Ma) ≈ 37 km Ma, or an equivalent of 37 mm a.

Case 3: Estimate the bounds if I don't know a reasonable average.
Some will be faster and some slower, so a value between say 10 and 100 mm a^{-1}. Many natural processes result in log-normally distributed values—especially those that vary by orders of magnitude—which means they are skewed towards lower values but have long tails on the high end. In such cases it makes more sense to use a geometric mean to calculate an average than an arithmetic mean. The geometric mean of 10 and 100 is simply the square root of their product, in our plate velocity example,

$$v_g = \sqrt{(10 \text{ mm a}^{-1})(100 \text{ mm a}^{-1})} \qquad = \sqrt{(1000 \text{ mm a}^{-1})} \approx 30 \text{ mm a}^{-1},$$

which in this case turns out to be about right.

0.3 Scientific notation

Scientific notation converts every number into a decimal multiplied by a power of 10, i.e., $m \times 10^n$, where m is a decimal value and n is an integer. The number m is called the mantissa or significand and n is called the order or exponent. When multiplying two numbers in scientific notation, we multiply the mantissa and add the orders,

$$(m_1 \times 10^{n_1})(m_2 \times 10^{n_2}) = (m_1 \cdot m_2) \times 10^{n_1+n_2}.$$

Division likewise is easier,

$$\frac{m_1 \times 10^{n_1}}{m_2 \times 10^{n_2}} = \frac{m_1}{m_2} \times 10^{n_1-n_2}.$$

Because we often only care about order-of-magnitude rather than precise estimates, we can limit our m_1 and m_2 to one to two digits.

Many unit conversion become a sinch in scientific notation. Using our seconds in a year above, if we needed the number of seconds in a million years (as geoscientists we often work in deep time) we multiply by 10^6. Hence, $(3 \times 10^7 \text{ s a}^{-1})(10^6 \text{ a Ma}^{-1}) = 3 \times 10^{13} \text{ s Ma}^{-1}$.

0.4 Dimensional analysis as a test

Dimensional analysis is the process of checking units during mathematical operations to ensure that the correct units are determined in the end. For example, if you calculate the force of gravity from $F = mg$, how do you know that you end up with the appropriate unit of newtons (N)? We can write this as an equation:

$$\begin{aligned} F &= mg \\ [\mathrm{N}] &\stackrel{?}{=} [\mathrm{kg}]\,[\mathrm{m\,s^{-2}}] \\ [\mathrm{N}] &= [\mathrm{kg\,m\,s^{-2}}] \end{aligned}$$

Of course, this can only be verified as correct if you know that a newton is defined as a $\mathrm{kg\,m\,s^{-2}}$, so it is useful to know how units are defined in terms of base units like m, s, kg, C, etc.

Let's look at another example, the calculation of pressure. Such a calculation is commonplace for geoscientist. Often we report pressures in mega-pascals, MPa. The equation for pressure is simple,

$$P = \rho g h,$$

where ρ is the average density of rock above the point of interest, g is the gravitational acceleration, and h, is the thickness of rock above. Density is typically reported in $\mathrm{kg\,m^{-3}}$, gravity in $\mathrm{m\,s^{-2}}$, and height in km. Note neither pressure or height are reported in standard SI units. So let's see if this works dimensionally,

$$\begin{aligned} [\mathrm{MPa}] &\stackrel{?}{=} [\mathrm{kg\,m^{-3}}]\,[\mathrm{m\,s^{-2}}]\,[\mathrm{km}] \\ [\mathrm{MPa}]\,[\mathrm{Pa\,MPa^{-1}} \cdot 10^{-6}] &\stackrel{?}{=} [\mathrm{kg\,m^{-2}\,s^{-2}}]\,[\mathrm{km}]\,[\mathrm{m\,km^{-1}} \cdot 10^{-3}] \\ 10^{-6}[\mathrm{Pa}] &\stackrel{?}{=} 10^{-3}[\mathrm{kg\,m^{-1}\,s^{-2}}], \end{aligned}$$

since a Pa is a $\mathrm{N\,m^{-2}}$ or a $\mathrm{kg\,s^{-2}\,m^{-1}}$, we can see

$$10^{-6}[\mathrm{Pa}] \neq 10^{-3}[\mathrm{Pa}].$$

So, we are missing a conversion factor of 10^{-3} to give our result with units in MPa.

Question 1. Tectonic Plate versus Freight Train
For this problem, I want you to answer two questions:

(a) Which has larger momentum? Momentum is given by $p = mv$, where m is mass and v is velocity.

(b) Which has larger kinetic energy? Kinetic energy is given by $K = \frac{1}{2}mv^2$.

Remember to work backwards. Start with mass, how do you estimate it for a tectonic plate, a freight train? Google won't be much help because it cannot tell you the answer (and often gives one orders of magnitude off when students ask). You'll need to do some out of the box thinking, recall some things you might know and some things you might not, for instance, what comprises a plate, not just horizontally, but vertically?

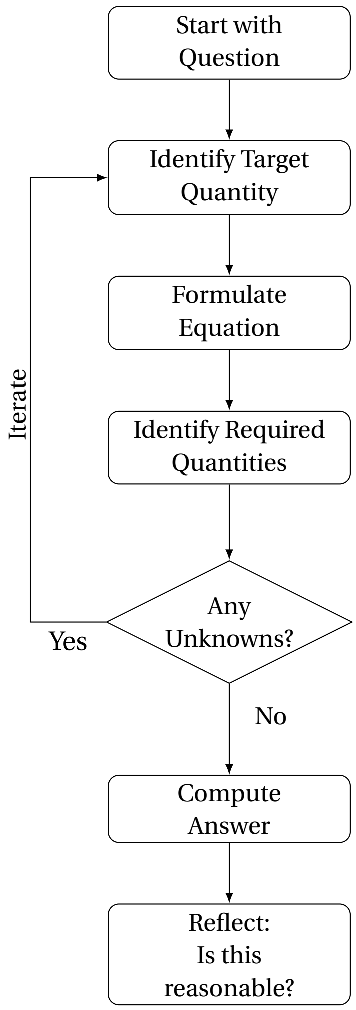
You can use Google for some basic geology questions, but don't ask it for numbers/quantities.

Use this page for tectonic plate.

Use this page for freight train plate.

Key Ideas

Order-of-magnitude estimation is a core scientific skill. The goal is not a precise answer but a defensible estimate within a factor of a few. By breaking a problem into components you *can* estimate, you avoid the trap of thinking "I don't know the answer" when you actually know enough to constrain it.



Back-of-the-envelope calculations work backwards from the desired quantity. Start with the final equation, identify the required inputs, and then recursively estimate any unknown quantities until every term can be evaluated.

Estimates are built from physical reasoning. Unknown quantities can often be constrained using geometry, scaling relationships, analogy, bounds, known properties, and dimensional analysis. The goal is not precision but a defensible order-of-magnitude estimate.

Always reflect on the result. A calculation is not finished when the arithmetic is complete. Ask whether the answer is physically reasonable, whether the units are correct, and how sensitive the estimate is to the assumptions made.

Activity 1.B
Sheet Model of Potential Fields

Purpose

This short activity uses a physical sheet model to develop intuition for scalar fields, gradients, and curvature. By fixing boundary values and introducing localized perturbations, you will explore how fields respond to boundary conditions and internal sources, and how these effects relate to geophysical potential fields.

Learning Outcomes

By the end of this activity, you should be able to:

- Explain how boundary conditions control the shape of a scalar field in the interior.
- Identify regions of high gradient magnitude and strong curvature from the geometry of a surface.
- Describe qualitatively why observation distance smooths short-wavelength field features.
- Connect boundary conditions and localized sources to the concepts of Laplace and Poisson equations.

Setup

Work as a group around a stretched sheet. Some students will hold the edges, while others gently push or pull on the sheet from below or above.

Tasks and Prompts

Question 1. Boundary conditions (Dirichlet)
Hold the edges of the sheet at fixed but varying heights and pull it taut.

(a) How do the boundary values control the shape of the interior of the sheet?

(b) Are there any local maxima or minima within the interior?

Question 2. Sources and sinks (Poisson region)
Introduce localized perturbations by pushing up (source) or pulling down (sink) at points within the sheet.

(a) How does the interior of the sheet respond to these perturbations?

(b) Do these changes affect the boundary? Why or why not?

Question 3. Gradients and curvature
Consider the geometry of the sheet as a surface.

(a) Where is $|\nabla\Phi|$ largest? How can you tell from the shape of the sheet?

(b) Where is $\nabla^2\Phi \neq 0$ versus ≈ 0?

(c) Which features correspond to regions of strong curvature?

Question 4. Effect of observation distance
Imagine observing the sheet from farther above.

(a) Why does the surface appear smoother as the observation distance increases?

(b) Which features disappear first, and why?

Key Ideas

Gradients and Curvature: Gradients describe how quickly and in what direction a field changes, while curvature describes how those changes themselves vary in space. In geophysical potential fields, boundary conditions and internal sources control the field, and increasing observation distance smooths short-wavelength features.

Laplace vs. Poisson: Where there are no sources or sinks, $\nabla^2\Phi = 0$ (Laplace equation) and the field is smooth with no interior extrema. Where sources are present, $\nabla^2\Phi \neq 0$ (Poisson equation) and local curvature appears.

Upward continuation: Moving the observation point away from the sources is equivalent to low-pass filtering the field—a concept we will revisit formally with Fourier methods in Week 7.

Activity 1.C
From Elevation to Gradients

"I will never listen to ocean waves or view a beautiful sunset in quite the same way again. That is perhaps the greatest gift one can gain by delving into calculus: It is a whole new way of looking at the world, accessible only through the realm of mathematics."

— Jennifer Ouellette, *The Calculus Diaries: How Math Can Help You Lose Weight, Win in Vegas, and Survive a Zombie Apocalypse*

Purpose

This activity develops an intuitive and mathematical understanding of gradients using familiar topographic examples. By sketching gradient vectors on contour maps and working through the underlying mathematics, you will connect the physical idea of slope to the formal definition of the gradient of a scalar field.

Learning Outcomes

By the end of this activity, you should be able to:

- Draw gradient vectors on a topographic contour map, indicating both direction (up slope) and relative magnitude (arrow length).
- Explain why the gradient is a vector quantity requiring both magnitude and direction.
- Compute the partial derivatives $\partial h/\partial x$ and $\partial h/\partial y$ from a discrete elevation profile and combine them to find the resultant gradient vector.
- Identify locations where the gradient magnitude is large or zero from the geometry of a topographic surface.
- Distinguish between the gradient in the x-direction, the y-direction, and the overall maximum slope direction.

Introduction

Geophysical measurements are fundamentally measurements of how physical fields vary in space. If gravity, temperature, magnetic intensity, or topography were constant everywhere, there would be little information from which to infer Earth structure or composition. The spatial variation of a field often contains more information than the field value itself.

One of the simplest ways to describe spatial variation is through the gradient. A gradient measures both the direction in which a field changes most rapidly and the rate at which that change occurs. Although gradients are used throughout geophysics, elevation provides a familiar and intuitive example because we can directly observe and experience slope.

In this activity you will explore gradients graphically using contour maps and mathematically using partial derivatives. The goal is to build intuition that will later be applied to more abstract fields such as gravity, magnetics, and temperature.

Figure 1: Mount Shasta, California (image from Google Earth).

Mount Shasta Example

Elevation is a good place to start to think about gradients because we all intuitively know what a slope means in this context. As far as topography goes, Mount Shasta is pretty simple. It is a stratovolcano at the southernmost end of the associated with the Cascade Volcanic Arc, and looks like a textbook example of a volcano in that it is fairly symmetric radially and sits in the middle of a vast flat plain.

Question 1. Shasta gradient

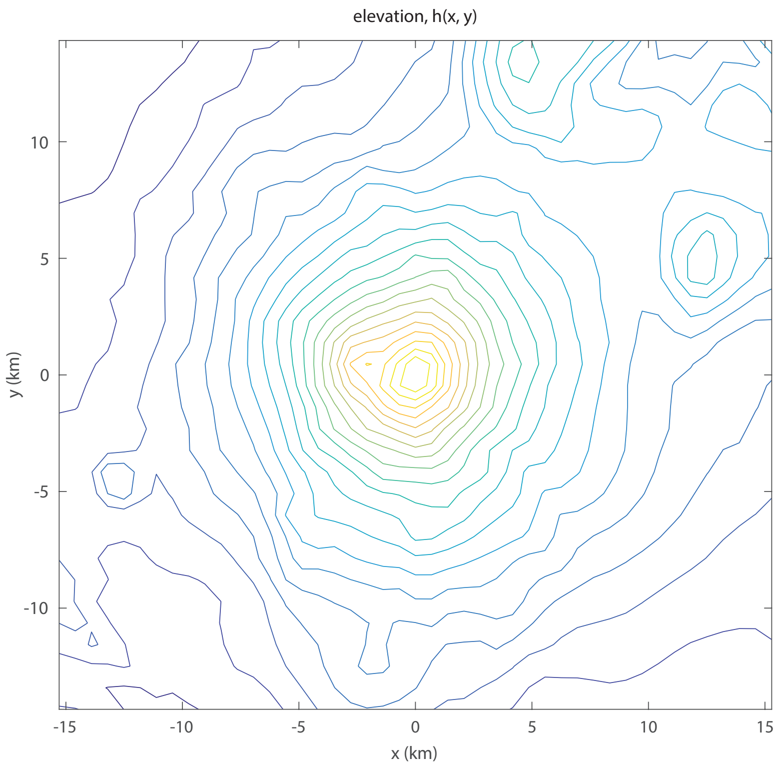


Figure 2: Topographic map of Mount Shasta.

(a) Over the topographic contours on Figure 2, draw a series of arrows pointing up slope representing the opposite direction water might flow. Make the arrows longer when the slope is steeper and make the arrows shorter when the slope is gentler.

(b) How did you decide to draw the direction of the arrows?

(c) How did you decide to draw the length of the arrows?

(d) Why does it require two quantities, length and direction, to describe the gradient?

The mathematics of gradients: the slopes of Mt. Shasta

Consider hiking a profile line over the peak of Mt. Shasta (Figure 3, top panel). You will start by walking up the slope of Mt. Shasta. The slope starts gently going up in elevation (a positive slope in Figure 3, bottom panel), which increases as you climb in elevation. The slope increases and reaches a maximum near the peak, then the slope starts to decrease as elevation still increases. Once you reach the snow-capped peak there is no change in slope (i.e., slope = 0). Then you decide to pull out your sled and quickly decend the other side (negative slope).

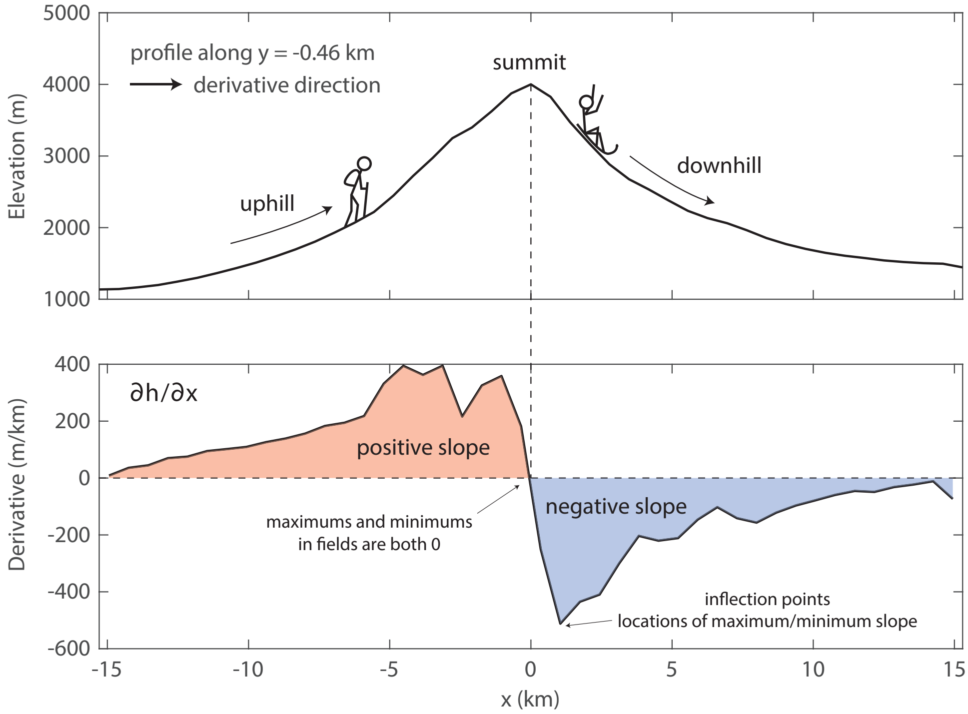


Figure 3: Elevation profile (top) and gradient along the elevation profile (bottom).

What we want to do is be able to describe the changes in slope that you as a hiker feels mathematically. Say every step you take is a similar distance laterally (the run) and the height you go up (or down, negative up) in elevation (the rise) allows us to express the slope as

$$\text{slope} = \frac{\text{rise}}{\text{run}}. \tag{1}$$

But the direction that you take matters. Since we don't necessarily know the direction of the maximum slope we'll force the hiker to walk in just one direction, say west to east ($+x$) to start. The size of the steps they take are Δx. In this case, we can express the slope as

$$\frac{\partial h}{\partial x} \approx s_x = \frac{h(x+\Delta x) - h(x)}{(x+\Delta x) - x} = \frac{h(x+\Delta x) - h(x)}{\Delta x}. \tag{2}$$

At this point the way we solve this mathematically as opposed to a computer are different. A computer will solve the problem much like above, taking the difference between the heights at $h(x+\Delta x)$ and $h(x)$ and dividing by the distance of each step (Δx).

For our purposes generally during the course, we will take a more mathematical approach, taking the case where we want to know the slope at a continous series of point along the elevation profile. In this case the distances between our steps approach zero and the slope we approximate the approaches a limit of the true slope at each point. This is the definition of the derivative, and you can find an extended discussion as a tutorial online and in the math appendicies in the course notes.

If instead, you were to hike from south to north ($+y$-direction) we we can write the slope as

$$\frac{\partial h}{\partial y} \approx s_y = \frac{h(y+\Delta y) - h(y)}{(y+\Delta y) - y} = \frac{h(y+\Delta y) - h(y)}{\Delta y}. \tag{3}$$

If you were to walk in a series of profiles in x or y, you could map out the slope with respect to that series of traverses. You will notice that the slope you get is very different depending upon whether you walked in the x-direction or the y-direction [1]. We can write the function of the slope, s at each location (x, y) as

$$\mathbf{s}(x, y) = s_x\hat{\mathbf{x}} + s_y\hat{\mathbf{y}}, \tag{4}$$

where the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are a unit vectors in the x- and y-directions, respectively. Note that the slope is a vector quantity, which means it has both a magnitude and a direction. In calculus, we describe the gradient ($\boldsymbol{\nabla}$) of the elevation function, $h(x, y)$, by

$$\boldsymbol{\nabla} h = \frac{\partial h}{\partial x}\hat{\mathbf{x}} + \frac{\partial h}{\partial y}\hat{\mathbf{y}}. \tag{5}$$

Because the components are in different directions, we cannot simply add them to find the maximum slope.

From the result above, we learn that the slope is different depending upon the direction you walk (N-S) versus (E-W) which confounds the direction of the maximum slope. You may also note that if we had walked in the opposite (negative) directions, we would have changed the sign of our slopes. Do not fear, there is a simple mathematical solution to determine both the maximum slope and its direction.

If we assume that from each point that we walked in either the +x or +y direction we can describe the local surface as a plane, the directions we walked were purposfully perpendicular to each other. It turns out that these two directions can be used to define that plane and both the angle and magnitude of the maximum slope (magnitude) along it. To determine the maximum slope along each individual little plane, we use the

[1] Since Mt. Shasta is fairly radially symmetric, you will notice that the results are pretty similar if you rotate the one of the plots 90° with respect to the other. This is a relatively special case as we can't generally do this, however, many of the problems we will solve in this course either change in one direction only or are symmetric in some way to make the problems simpler.

Pythagorean theorem, which you may remember from high school,

$$\|s\| = (s_x^2 + s_y^2)^{1/2}. \tag{6}$$

Note that because we square each slope component before adding them, we have ensured that our resultant magniude of the slope is positive. To determine the direction of the maximum slope, we can use trigonometry to determine the angle with respect to the x-axis as

$$\tan\theta = \frac{s_y}{s_x}. \tag{7}$$

Note that to find the appropriate angle we will have to keep track of the signs of s_x and s_y to ensure that θ lies in the appropriate quadrant.

Marble Canyon Example

Figure 4: Marble Canyon, Arizona (image from Google Earth).

Let's look at another topographic example at Marble Canyon in Arizona. Marble Canyon sits before the entrance of the Grand Canyon. The area surrounding the canyon has relatively flat topography except for some steep-sided cliffs, one east of the canyon that runs approximately north to south and a second, north of the canyon which varies in azimuth. The canyon itself runs northeast to southwest though the center of the image, carved by the Colorado River.

In this exercise, we will take the previous example one step further by determining the slope along each individual axis and then combine them as we might in the mathematical sense and then combine the results to obtain determine a magnitude and direction. Again, we'll do this graphically rather than mathematically to gain further insight into how the mathematics works. Note there is a substantial region where the topography is relatively flat (i.e., the slope is negligible). You don't need to put any arrows in these positions so focus on regions where elevation changes.

Question 2. Marble Canyon

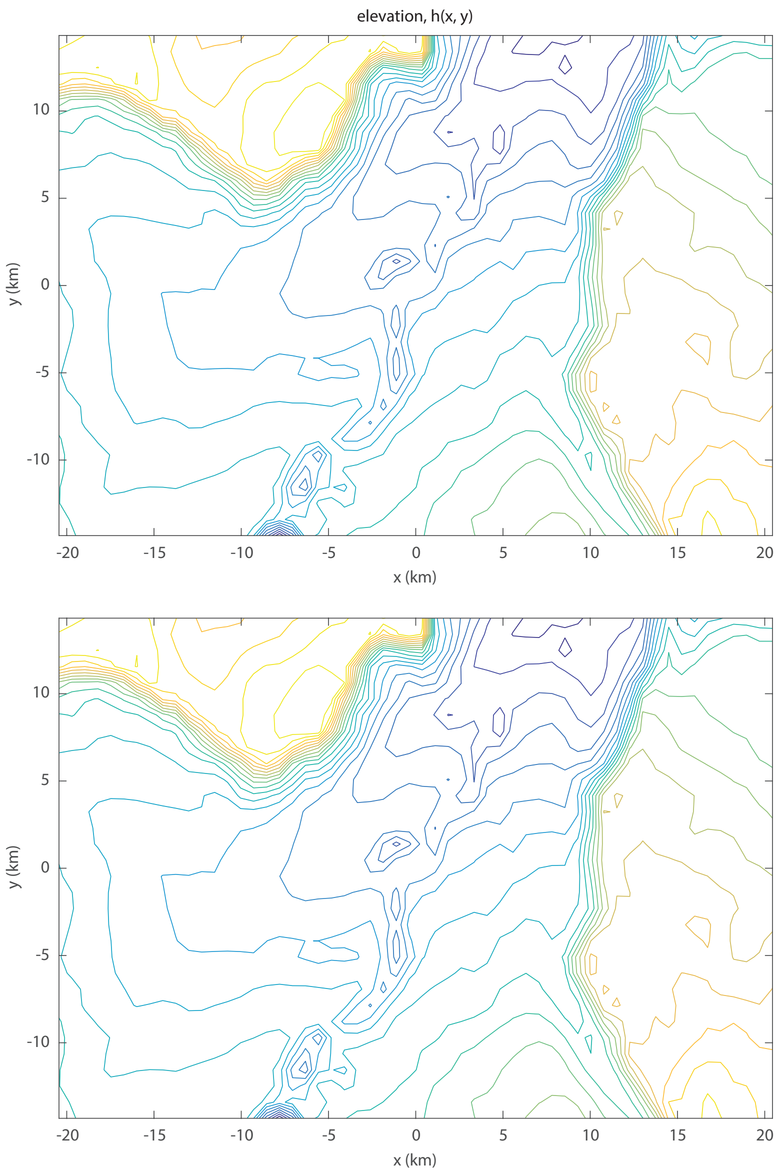


Figure 5: Topographic map of Marble Canyon.

(a) In Figure 5 are two of the same topographic map of Marble Canyon. On the top map, make a series of arrows parallel to the x-axis with length reprepresenting the slope in the x-direction. Again, arrow should point up slope. Now on the same topographic map, make a series of arrows parallel to the y-axis with length representing the slope in the y-direction. Once you have both sets of arrows, create a new set of arrows on the bottom topograpic map that represent a combination of the two. The resultant vector is simply a graphical application of the Pythagorean equation. What you will have done is the same thing that the math is doing. So when we are working with abstract fields that cannot be seen or felt by us, remember, the concept is shown here is the same.

(b) How do the individual derivative components behave with respect to the ridge running north east of the canyon?

(c) For the northern ridge, why does the y-component have only positive slopes whereas the x-component is positive in some places and negative in others?

(d) Can you find examples of either of the above situations occuring in the canyon itself?

Key Ideas

The gradient is a vector field: it has both a magnitude (the steepness of the slope) and a direction (pointing up the steepest ascent). Two numbers — the partial derivatives in x and y — are required to fully describe the gradient at any point.

Direction matters: the slope you measure depends on the direction you walk. The maximum slope is found by combining the two perpendicular components using the Pythagorean theorem: $|\nabla h| = \sqrt{(\partial h/\partial x)^2 + (\partial h/\partial y)^2}$.

Contour maps and gradients: the gradient is always perpendicular to contour lines and points toward higher values. Where contours are closely spaced, the gradient magnitude is large; where they are widely spaced, the gradient is small.

From elevation to geophysics: the same mathematical machinery — partial derivatives, vector combination, magnitude and direction — describes the gradient of any scalar field, including gravity, magnetic, and temperature fields.